

Sharif University of Technology

Department of Electrical Engineering

Research Proposal

Smoke and Fire Detection Project Using CCTV Cameras

This proposal depicts a formal research plan for developing an early smoke and fire detection framework using CCTV imagery, hybrid dataset engineering, YOLO-based detection, and temporal verification for continuous monitoring in industrial environments.

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Abstract

Early fire detection in industrial environments is crucial for preventing irreversible financial losses and threats to human life. In this work, an intelligent model based on Convolutional Neural Networks (CNN) is designed for accurate, real-time detection of smoke and fire. To achieve an efficient trade-off between inference speed and detection accuracy, the advanced YOLOv11 and YOLOv9 architectures are used as the core of the network. Given the challenge of limited high-quality labeled data, a comprehensive, industrial-oriented dataset was developed. This dataset is hybrid: part of it consists of available real-world videos, while the remaining part is composed of synthetic images generated by Generative AI models. Using synthetic data increases scenario diversity and improves the model's generalization ability in recognizing different smoke patterns. Finally, the proposed model was evaluated on diverse test sets; the results indicate high accuracy and reliable performance under varying environmental conditions.

Keywords: Smoke and fire detection, deep learning, YOLOv11, YOLOv9, convolutional neural networks dataset

1. Introduction

Early fire detection remains one of the most important safety requirements in industrial halls, warehouses, sheds, and workshops because even a short delay can lead to production downtime, equipment damage, financial loss, and threats to human life. In conventional practice, smoke, heat, or flame sensors are installed at fixed points and alarms are triggered when physical cues reach those sensors. Although these systems are widely used, their effectiveness can be limited in high-ceiling spaces, dusty sites, strongly ventilated areas, and large environments in which smoke needs time to propagate upward.

At the same time, many industrial sites already contain CCTV infrastructure for surveillance and process monitoring. This creates an opportunity to use existing cameras as visual sensing devices that detect smoke and fire closer to the ignition source and over a broader spatial field. The source report emphasizes that this approach is attractive not only because it can shorten the response window, but also because it can reduce the need for dense additional hardware deployment in some monitored areas [1, 2].

Despite that promise, smoke is difficult to model visually. Its transparency, texture, scale, and motion vary substantially with lighting, background, distance, and airflow. Furthermore, steam, fog, dust, reflections, and shadows can appear visually similar to smoke. For this reason, a reliable system must go beyond single-frame appearance and combine robust object detection, careful dataset engineering, and video-level decision logic.

2. Problem Statement

The central problem of this proposal is the lack of a practical early smoke and fire detection framework that can operate on CCTV video streams in industrial environments with acceptable speed, accuracy, and resistance to false alarms.

This is a meaningful problem for three reasons. First, ceiling-mounted point sensors may respond late when smoke begins near the ground, behind machinery, or far from the sensor location. Second, image-based systems often fail in real settings because early smoke has ambiguous boundaries and because non-smoke phenomena can trigger visually plausible false positives. Third, many learning-based solutions are constrained by limited domain-aligned training data, especially for faint smoke, CCTV viewpoints, and rare yet safety-critical operating conditions.

The missing piece is not only a detector architecture. The main gap is the absence of an integrated framework that jointly addresses early-stage visibility, dataset coverage, and temporal robustness under continuous monitoring. This proposal therefore seeks to design a unified pipeline that combines data-centric development, real-time detection, and

multi-frame verification.

3. Research Objectives

The proposed work has four measurable objectives:

1. To construct a hybrid dataset that combines real CCTV-like smoke and fire imagery with controlled synthetic samples covering thin smoke, diverse industrial backgrounds, and difficult negative cases.
2. To design and train a YOLO-family detector for real-time smoke and fire localization under varying lighting, distance, and viewpoint conditions.
3. To develop a temporal verification mechanism that analyzes multi-frame detection behavior and reduces false alarms caused by static or quasi-static smoke-like patterns.
4. To evaluate the full framework using accuracy, robustness, latency, and deployment-oriented performance measures, and to identify a configuration suitable for practical use.

4. Scope of Study

4.1. Included

The scope includes industrial and semi-industrial CCTV scenarios, RGB image and video streams from fixed surveillance cameras, early smoke and visible fire events, and challenging negative cases such as steam, fog, dust, cloud-like regions, and lighting artifacts. The study also includes detector training, temporal verification, and deployment-oriented evaluation.

4.2. Not Included

The scope does not include wildfire forecasting, drone-only monitoring, mandatory thermal or multispectral sensing, suppression-system actuation design, evacuation modeling, commercial certification, or full productization and long-term maintenance studies.

5. Literature Review

The technical foundation of this proposal follows the development of modern object detection. R-CNN demonstrated that deep convolutional features can be highly effective for object localization and classification, but its multi-stage region-proposal pipeline is computationally demanding and not naturally suited to real-time CCTV monitoring [2]. In contrast, YOLO formulates detection as a single-pass prediction problem and is therefore better aligned with low-latency surveillance applications [1].

For smoke and fire detection, the literature suggests that model selection alone is not sufficient. Performance is strongly tied to dataset quality, viewpoint alignment, and the ability to reject visually similar non-smoke patterns. This proposal adopts that perspective by treating data engineering and temporal decision logic as first-class design components rather than secondary implementation details.

6. Research Gap

Although previous studies show that deep learning can support visual smoke and fire detection, several practical issues remain insufficiently addressed.

First, many systems emphasize visible flame or dense smoke rather than faint early-stage smoke, even though early smoke detection is the main requirement for timely intervention. Second, public or easily available datasets often do not reflect industrial CCTV viewpoints, background diversity, and hard-negative conditions that occur in practice. Third, many reported results focus on frame-level detection, while continuous video operation requires an additional mechanism to distinguish true smoke dynamics from visually similar static objects and transient artifacts.

Accordingly, the research gap is the lack of a single framework that integrates targeted dataset enrichment, real-time YOLO-based detection, and temporal filtering specifically for early smoke and fire detection in industrial CCTV environments.

7. Methodology

7.1. Overall Framework

The proposed workflow is shown in Fig. 1. It begins with data acquisition and annotation, proceeds through hybrid dataset development and model training, and ends with temporal verification and deployment-oriented evaluation.

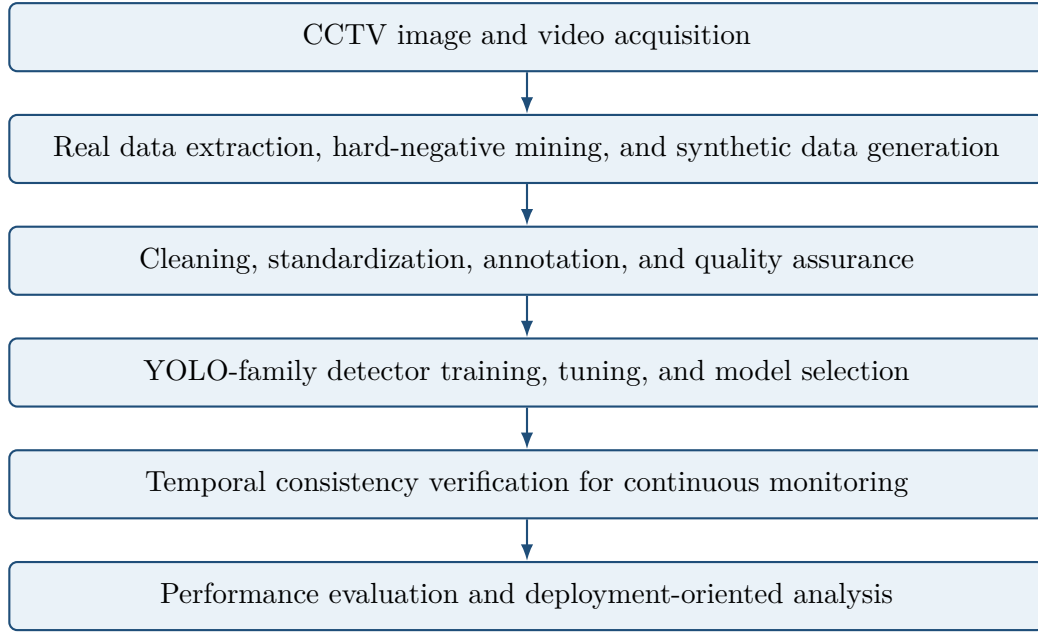


Figure 1: Overall methodological framework of the proposed study.

7.2. Data Collection

The dataset will be constructed from two complementary sources. The first source is real data: video frames and still images collected from CCTV-like viewpoints, industrial scenes, public monitoring footage, and curated visual material showing smoke or fire events. The second source is synthetic data generated specifically to cover rare or underrepresented cases, including thin smoke, low-light scenes, bright backgrounds, small smoke regions, and partially occluded smoke.

All usable samples will be cleaned, standardized, and manually annotated with bounding boxes. Special attention will be given to hard negatives such as steam, dust, fog, reflections, cloud-like structures, and contrast changes. The study does not involve human participants; the primary data source is visual surveillance content.

7.3. System Design

The detector will be chosen from the YOLO family because this class of models is designed for high-throughput object detection. The system will include three logical layers:

1. *Detection layer*: localization of smoke and fire in individual frames.
2. *Decision layer*: use of two confidence thresholds, where high-confidence detections are accepted directly, low-confidence outputs are rejected, and intermediate-confidence cases are forwarded for temporal review.
3. *Temporal layer*: analysis of box position and size variation across consecutive frames to distinguish real smoke dynamics from visually similar static patterns.

Implementation will primarily use Python together with PyTorch, Ultralytics, OpenCV, and standard annotation tools.

7.4. Data Analysis

Evaluation will use training, validation, and test partitions that preserve scenario diversity. The main quantitative measures will be mAP@50, mAP@50:95, precision, recall, and F1-score. Because the end goal is practical deployment, the study will also report false-alarm rate, missed-detection rate, approximate frames-per-second throughput, and inference latency.

The experimental plan includes comparing baseline image-only detection against the full framework with temporal verification. Ablation experiments will analyze the effect of synthetic data ratio, hard-negative coverage, and threshold settings for temporal decision-making.

8. Preliminary Results

Although this document is written as a proposal, the source project already includes initial experiments that can be used as a baseline for the proposed study. Four training scenarios were evaluated on a test set with 42 images and 55 labels using mAP@50, mAP@50:95, precision, and recall. As summarized in Table 1, the YOLOv11-n configuration achieved the strongest overall trade-off, while the comparison between Scenarios 1 and 4 shows that data quality and training conditions can significantly affect final performance even when the architecture remains unchanged.

Table 1: Summary of preliminary evaluation results from the source report.

Scenario	Model	mAP50	mAP50–95	Precision	Recall
1	YOLOv11-n	83.0	39.0	81.0	71.0
2	YOLOv11-m	69.0	33.0	61.0	69.0
3	YOLOv9-e	78.0	33.0	82.0	68.0
4	YOLOv11-n	41.0	19.0	59.0	30.0

These preliminary results are useful for the proposal because they already indicate two practical lessons that motivate the planned research. First, a lightweight detector can outperform a larger alternative when the task involves thin smoke and limited domain-specific training data. Second, model performance depends strongly on dataset construction, cleaning, and alignment with CCTV operating conditions, which justifies the proposal’s emphasis on hybrid dataset engineering and temporal verification.

9. Expected Results and Contribution

The expected result is a deployable research prototype that improves the reliability of early smoke and fire detection in industrial CCTV environments. More specifically, the study is expected to contribute:

- a hybrid dataset design strategy that strengthens weakly represented early-smoke cases,
- a YOLO-based real-time detector tailored to CCTV operating conditions,
- a temporal verification method that reduces false alarms from smoke-like static patterns, and
- an evidence-based analysis of how detector size, data quality, and temporal logic interact in a deployment-oriented setting.

In practical terms, the proposed framework is expected to improve early warning performance and provide a software-based layer that complements conventional fire safety systems in monitored facilities.

10. Work Plan

The revised work plan follows the requested sequence of activities and durations. Table 2 presents the schedule in a concise narrative format, while Fig. ?? visualizes the same plan as a compact 22-week timeline.

Table 2: Planned research tasks and durations.

Task	Duration	Scheduled Period
Literature review	1 month	Weeks 1–4
Real data collection	2 weeks	Weeks 5–6
Synthetic data generation	2 weeks	Weeks 7–8
Annotation & quality control	1 week	Week 9
Detector training	3 weeks	Weeks 10–12
Evaluation & analysis	2 weeks	Weeks 13–14
New methods for increasing accuracy	1 month	Weeks 15–18
Writing & submission	1 month	Weeks 19–22

References

- [1] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” arXiv preprint arXiv:1506.02640, 2016.
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